

NIRANJAN SHRESTHA

Power Electronics Researcher - EV Charging Converters, Resonant Conversion & Control
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RESEARCH PROFILE

Power electronics researcher and design engineer with **6+ years developing high-efficiency power conversion systems for electric vehicle charging**, spanning topology research, dynamic modeling, closed-loop control design, embedded firmware, PCB design, and experimental validation. Led end-to-end research and development of a **3.7 kW EV charging platform** built on a novel resonant compensation network and phase-shift control scheme for CC/CV charging — an NSERC- and MITACS-funded program delivered with an industry partner. Author or co-author of nine publications (eight IEEE journal and conference papers, one book chapter), 50 citations, h-index 5, with recognition through the IEEE ITEC Student Travel Award.

Currently conducting applied research and development on novel system architecture, advanced converter topologies, and control methods for EV charging up to 7.7 kW at Wireless PnC Inc., with responsibility for identifying research directions, prototyping, hardware testing/validation mentoring junior engineers, and translating findings into manufacturable designs compliant with SAE, CSA, and UL automotive standards. Seeking to apply converter design and control research to the next generation of high-power EV supercharging infrastructure.

TECHNICAL & RESEARCH COMPETENCIES

Power Conversion Topologies: Resonant converters (LLC / series-parallel) · Boost PFC rectifiers · Full-bridge and half-bridge inverters · DC-DC converters · Synchronous and diode rectification · Compensation network design · Soft-switching (ZVS/ZCS) · High- and low-frequency converter design

Control Design: Phase-shift control · Frequency control · Constant-current / constant-voltage (CC/CV) charging control · Closed-loop feedback design · Small-signal modeling and stability analysis · Load-transient and misalignment-tolerant control

Circuit Design & Simulation: MATLAB Simulink/Simscape · PLECS · LTspice · Proteus Design Suite · M-language · Circuit- and system-level co-simulation · Model validation against experimental hardware

Closed-loop Control Design, Simulation & Validation: Closed-loop control validation · Automated code generation from models to real-time targets · Validation test plan authoring · Fault injection and root-cause analysis · OPAL-RT & RT-LAB

Embedded Systems & Firmware: TI C2000 microcontrollers · STM32 · C / C++ · Python · Digital control loop implementation · PWM and gate-drive control · SPI, UART communication

Hardware Realization: Extensive Hardware testing/validation experience · KiCad for schematic and PCB layout design · PCB fabrication and testing · Thermal and efficiency characterization · Bench and vehicle-level test execution · AutoCAD

Standards & Documentation: SAE, CSA, UL (wireless power transfer standards) · IEEE 519 harmonic compliance · BS 7671 electrical test and inspection · Technical specification authoring · Design documentation control · Patent-ready disclosure and conference/journal writing

EDUCATION

Master of Applied Science (MAsc), Electrical & Computer Engineering

Ontario Tech University, Oshawa, Ontario, Canada · Jan 2022 – Apr 2024

Research: High-efficiency 3.7 kW resonant wireless power transfer for EV charging - Novel compensation network topology and phase-shift control for CC/CV operation. Supervisor: Dr. Sheldon Williamson. Funding: NSERC, MITACS Accelerate (industry partner collaboration).

Bachelor of Engineering, Electrical Engineering

Tribhuvan University, Pulchowk Campus, Lalitpur, Nepal · Nov 2013 – Sep 2017

Focus: Power electronics, control systems.

RESEARCH & PROFESSIONAL EXPERIENCE

Electrical Engineer — Power Conversion and control R&D

Wireless PnC Inc. · Victoria, British Columbia (Hybrid) · May 2024 – Present

- Conduct applied research into **advanced converter topologies and control schemes for wireless charging**, identifying technical directions and evaluating emerging power conversion approaches against product roadmaps; **3** concepts advanced to working prototype for E-bike and E-scooter WPT system.
- Own end-to-end design, dynamic modeling, circuit- and system-level simulation, control design, and hardware validation of wireless charging system up to **7.7 kW** across EV platforms, delivering **95% dc-dc efficiency**.
- Design and fabricate power stages including **Boost PFC rectifiers and full-bridge inverters**, achieving **93 %** end-to-end efficiency for E-bike/E-scooter WPT system and **3.6% THD** (source current) under non-linear loading, evaluated against IEEE 519 and automotive power quality requirements.

- Author technical specifications, detailed schematics, and design documentation; perform **KiCad** schematic and layout design & fabrication, and hardware bring-up for **8** converter boards.
- Collaborate across hardware, firmware, and mechanical engineering to integrate converter subsystems into complete charging platforms, resolving cross-domain integration issues through joint root-cause analysis.
- Mentor and technically supervise **4** electrical interns across simulation, hardware design, and validation, establishing reusable modeling and test-procedure frameworks now applied across **7.7 kW WPT** projects.
- Author validation test plans for all project phases; execute bench and vehicle-level testing and report findings to engineering leadership.
- Maintain currency with evolving automotive charging and WPT standards (SAE, CSA, UL) and integrate requirements into design practice.

Graduate Research Assistant — Power Electronics and Control for Wireless charging system

Ontario Tech University · Oshawa, Ontario · Jan 2022 – Apr 2024

- Delivered complete R&D of a **high-efficiency resonant wireless charging system for EVs**, achieving **92.5%** peak DC-to-DC efficiency at **3.7 kW**, funded by **NSERC and MITACS** in collaboration with an industry partner.
- Researched and validated a **novel compensation network topology** paired with a phase-shift control scheme enabling seamless CC/CV charging transitions — the core thesis contribution and basis of **3 first-author IEEE** paper publications.
- Developed and experimentally verified dynamic models of DC–DC converters, inverters, rectifiers, and resonant converters in **MATLAB/Simulink, Simscape, and PLECS**, correlating simulated efficiency and transient response to measured hardware within **5%**.
- Executed **closed-loop control validation** by deploying **Simulink/Simscape models to embedded targets**, verifying control logic against hardware during system integration and testing.
- Characterized system performance across **3** coil geometries and **3** compensation configurations, quantifying efficiency and power-transfer sensitivity to misalignment and air-gap variation.
- Presented findings at IEEE conferences; awarded the **IEEE ITEC Student Travel Award (2023)** and placed **1st Runner-Up in the ITEC Coil Winding Competition (2023)**.

Electrical Engineering Intern — Distributed Generation

Oshawa Power · Oshawa, Ontario · Jun 2022 – Apr 2023

- Developed regression-based forecasting models for energy production and revenue across **5** solar PV and wind generation assets, producing capacity and yield analyses used in utility planning.
- Prepared micro-embedded generation connection applications for net-metering distributed energy resource interconnections.
- Audited vendor equipment test reports against applicable industry standards, identifying non-conformances prior to approval.

Project Engineer — Solar PV System Design

Sunshine Energy Pvt. Ltd. · Kathmandu, Nepal · Nov 2019 – Dec 2021

- Delivered system design, sizing calculations, single-line diagrams, cost estimation, and tender bid documentation for **15** off-grid and on-grid solar PV, solar water pumping system, solar home, and street lighting projects totaling nearly **60 kW**.
- Conducted site surveys and translated field constraints into generation capacity, storage sizing, and runtime design decisions.

Electronics Technician — Electrical Test & Maintenance

Defence Infrastructure Organisation – British Gurkhas Nepal · Lalitpur, Nepal · Feb 2018 – Aug 2019

- Diagnosed and repaired faults on generators, motors, boilers, and solar thermal systems along with their necessary maintenance and servicing.
- Executed electrical installation testing and inspection to **BS 7671:2008**, producing compliant certification documentation.

PUBLICATIONS & RESEARCH OUTPUT

9 peer-reviewed publications · 50 citations · h-index 5

Journal Article

- Y. Shanmugam, R. Narayanamoorthi, V. K. Ramachandaramurthy, P. Bernat, *et al.*, “Machine learning based optimal design of on-road charging lane for smart cities applications,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 4, 2024.

Book Chapter

- **N. Shrestha**, J. Son, and S. Williamson, “Smart charging of electric vehicles,” in *Battery Technologies for E-Mobility*, 2025, pp. 289–337.

Conference Publications

- **N. Shrestha**, V. S. R. V. Oruganti, and S. Williamson, “Digitally controlled misalignment-tolerant inductive power transfer system with adaptive hybrid compensation for CC/CV charging of e-scooter,” in *Proc. 2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2025, pp. 801–808.

- **N. Shrestha** and S. Williamson, "Hybrid compensation network for misalignment-tolerant constant current/constant voltage charging for wireless power transfer system," in *Proc. 2024 IEEE Transportation Electrification Conference and Expo (ITEC)*, 2024, pp. 1–6.
- J. Son, **N. Shrestha**, and S. Williamson, "Comprehensive comparative analysis of circular, rectangular, and hexagonal coils for wireless charging of e-mobility," in *Proc. 2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2024, pp. 2915–2921.
- **N. Shrestha**, J. Son, and S. Williamson, "Constant current/constant voltage charging via series-series compensated resonant inductive wireless charging for electric vehicle," in *Proc. 2023 IEEE Transportation Electrification Conference and Expo (ITEC)*, 2023, pp. 1–6.
- **N. Shrestha**, A. Samanta, F. C. Fietosa, and S. Williamson, "State-of-the-art wireless charging systems for e-bikes: Technologies and applications," in *Proc. 2023 IEEE 14th International Conference on Power Electronics and Drive Systems (PEDS)*, 2023.
- A. Samanta, A. Huynh, **N. Shrestha**, and S. Williamson, "Combined data driven and online impedance measurement-based lithium-ion battery state of health estimation for electric vehicle battery management systems," in *Proc. 2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2023, pp. 862–866.
- K. T. Lulbadda, R. De Seram, **N. Shrestha**, T. Sidhu, and S. S. Williamson, "An analysis of advanced soft-switching techniques for dc–ac power converters based on auxiliary circuits," in *Proc. 2023 IEEE International Conference on Industrial Technology (ICIT)*, 2023, pp. 1–6.

Invited Tutorial at WPTCE Conference

- S. Williamson, V. Oruganti, and **N. Shrestha**, "Next-Generation Wireless Charging for EVs and Micromobility: Advanced Couplers with Adaptive Compensation-based Control and SiC Converter-based Charging Architectures with V2G Capability," *2026 IEEE Wireless Power Technology Conference and Expo (WPTCE)*, Halifax, NS, Canada, July 2026.

Thesis

- **N. Shrestha**, "A hybrid compensation-based misalignment tolerant wireless power transfer system for e-mobility," M.A.Sc. thesis, Ontario Tech University, Oshawa, ON, Canada, 2024.

AWARDS & PROFESSIONAL ACTIVITY

- **IEEE ITEC Student Travel Award** — 2023
- **IEEE ITEC Coil Winding Competition, 1st Runner-Up** — 2023
- **IEEE SPEC Scholarship Award** — 2022
- **IEEE Toronto, Durham Region Sub-section** — Active Volunteer, May 2022 – April 2024
- Event Organizer, Energy Hackathon, Lalitpur, Nepal — 2017

REFERENCE

Available upon request